Asael Acosta

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EDUCATION

Massachusetts Institute of Technology, Cambridge, Massachusetts — May 2028

Candidate for B.S. in Aerospace Engineering — GPA: 4.3 / 5.0

Coursework*: Multivariable Calculus, Differential Equations, Discrete Math, Signals and Systems, Thermodynamics, Fluid Dynamics, Astrodynamics, Space Propulsion*

SKILLS

Programming: Python (PyTorch, Orekit, SciPy), LaTeX, Bash, Java (WPILib), /HTML/CSS/ JS (jQuery), MATLAB
Software: QGIS, Adobe Creative Suite (Photoshop, Lightroom, Illustrator), Miro, Microsoft Suite, Ansys STK (Astrogator), nTop, Solidworks, Git
Languages: English (native), Spanish (fluent)

EXPERIENCE

MITRE Corporation, Systems/Digital Engineering Intern Bedford, MA — June - August 2025

- Developing systems analysis tool for government sponsor, leveraging git for collaboration
- Meeting regularly with sponsor representatives to clarify project requirements and add features
- Developing project management tool using Microsoft Suite to streamline collaboration between sponsor and John Hopkins Applied Physics Laboratory
- Outlining robust data pipelines between information sources and displays to increase productivity metrics

MIT Satellite Club, *Data and Communications Subteams*, Cambridge, MA — October 2025 - Present

- Iterating on NASA Human Lander Challenge finalist publication in preparation for launching it in a partnered nanosatellite mission set to launch summer 2026
- Acquired amateur radio license for ground station work for May high altitude balloon launch
- Designing and testing satellite subsystems on ground before placing them on payload
- Practicing mission analysis, stakeholder satisfaction, and developing competence in industry software
- Presenting Concept of Operations, Preliminary Design review to visiting industry representatives

MIT Varon Lab, *Undergraduate Researcher*, Cambridge, MA — September 2025 - May 2026

- Training U-Net model to predict methane-sensitive bands from non-sensitive satellite imagery to improve methane point-source and plume detection
- Collecting and preprocessing ABI GOES data for training, validation, and testing datasets
- Investigating novel ML approaches used by the group and presenting progress at weekly meetings

National Polytechnic Institute, *AI Research Intern*, Querétaro, Mexico — May 2025 - August 2025

- Integrated Google DeepMind's Gencast forecasting model into a locally hosted hurricane trajectory predictor and visualization tool, reducing blurring from its previous iteration
- Translated and summarized ~20 English-written papers on AI weather forecasting techniques onto Spanish LaTeX document for future work on forecast downscaling
- Presented research to CENAPRED, the Mexican disaster prevention and response bureau, and to Google DeepMind representative in discussing further collaboration in their disaster prevention and mitigation initiatives

PROJECTS

NASA RASC-AL Challenge (MELIORA), *Concept of Operations*, Cambridge, MA — October 2025 - Present

- Selected as finalist and later one of the overall RASC-AL forum winners
- Iterating over final submission to submit for SciTech presentation in January, 2027
- Designing system architecture for communication, navigation, positioning, and timing for Mars mission concepts
- Synthesizing research on industry-standard communication systems to support preliminary design decisions
- Working with cross-functional subteams to define system requirements and stakeholder needs

Astrodynamics, *SSO Mission Planning and Analysis*, Cambridge, MA — February 2026 - Present

- Simulating SSO orbit under various degrees of perturbation, and using the results to inform thrust maneuver strategies to maximize mission duration and gracefulness of degradation
- Presenting findings in written report and final presentation given to class and industry representatives
- Term project for graduate coursework, done under guidance of professor and NASA JPL engineers